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Encoder-based Transformers

> The unsung heroes of AI.
>
> Built for deep understanding, not writing.

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THE SPLIT

The original 2017 Transformer had two distinct halves. Over time, the AI industry split them up for different jobs.

1. The Encoder (The Reader)

Takes in text, analyzes the deep context, and converts it into mathematical representations.

2. The Decoder (The Writer)

Takes those representations and generates new text, one word at a time.

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THE SUPERPOWER

Bi-directional Context

Humans read left-to-right. Decoders generate left-to-right. But **Encoders read in both directions at once.**

When an Encoder looks at the word "Apple", it simultaneously looks at all the words *before* it AND all the words *after* it to perfectly deduce the meaning.

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WHY DECODERS CAN'T DO THIS

The Blindfold

Generative models (like GPT) are "masked". When predicting word #5, they are legally not allowed to look at word #6.

Encoders have no blindfolds.

Because Encoders aren't trying to generate the future, they are allowed to see the *entire* document instantly. This results in far deeper comprehension.

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THE KING OF ENCODERS

BERT

(2018, Google)

Bidirectional Encoder Representations from Transformers.

> While OpenAI was building GPT to write essays, Google built BERT to master the exact intent behind human search queries.

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TRAINING METHOD

Fill in the Blanks

Masked Language Modeling

How do you train an AI to understand context without generating text? You play Mad Libs.

The Training Process:

Take a million books. Randomly black out 15% of the words.

"The quick brown [MASK] jumps over the lazy dog."

Force the AI to guess the masked word based on the surrounding text.

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WHAT ARE THEY GOOD AT?

CLASSIFICATION

Sentiment Analysis & Spam Detection.

Reading an email and deciding: "Is this positive, negative, or spam?"

SEMANTIC SEARCH

Matching questions to answers.

Turning documents into mathematical embeddings so a search engine can find them instantly.

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REAL WORLD IMPACT

Saving Google

In 2019, Google integrated BERT into its core search engine, causing the biggest leap in search quality in 5 years.

Query: "can you get medicine **for someone** pharmacy"

Before BERT, Google ignored "for someone" and returned results on how to get a prescription filled. BERT understood the nuance of picking up medicine on behalf of someone else.

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THE CHEAT SHEET

Encoder-Only (BERT)

Reads bidirectionally. Best for understanding, classifying, and extracting information from text.

Decoder-Only (GPT)

Reads unidirectionally. Best for generating new text, writing essays, and holding conversations.

Encoder-Decoder (T5 / BART)

Uses both. Best for translation and summarization (read the whole input, write a new output).

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Encoder Models

CONCEPT UNLOCKED

- 🪞 Reads Bi-directionally
- 🤖 The brain behind BERT
- 🔎 Revolutionized modern Search
- 📊 Masters Classification & Analysis

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